

## RESCUE JUNIOR - Bill of Materials (BOM)

Team name: Tuton

Every Line/Maze team has to submit a bill of materials for their robot

### Instructions:

|                                                                                                                                                                                                                                                         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| *Enter Team name above                                                                                                                                                                                                                                  |
| *All costs need to be in local currency and their approximate conversion into dollars.                                                                                                                                                                  |
| *All components worth less can be summarized in one line quantity 1 and their total cost (e.g. Screws)                                                                                                                                                  |
| *In the Local Currency Name column, the team must enter the name of the local currency, if it is different from US dollars.                                                                                                                             |
| *In the software sheet, in the Author column, the name of the designer, writer, or company that owns the software or library should be entered. If the team created their own algorithm, dataset, or AI model, they can include their names as authors. |
| *On the hardware sheet, in the "Author" column, enter the name of the company, engineer, or part manufacturer. If the team made a custom part, such as the robot housing, the team should include their names as the authors.                           |

**\*IMPORTANT\*** The "Hardware" sheet is not required for simulation

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| Name of the Local Currency | Total Cost Local Currency | Total Cost (\$) U.S.A. Dollars |
|----------------------------|---------------------------|--------------------------------|
| Japanese Yen JPY           | ¥126,514                  | \$784.39                       |

| #  | Component                      | Part name                  | Author                                   | Source                                                                                                  | Quantity | Unit cost (Local Currency) | Unit cost (US Dollars) | (Local  | (USA Dollars) |
|----|--------------------------------|----------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------|----------|----------------------------|------------------------|---------|---------------|
| 1  | PCB with electronic components | Custom made                | Shuji Igarashi / JLCPCB                  | <a href="https://jlcpcb.com/">https://jlcpcb.com/</a>                                                   | 1        | ¥6,000                     | \$37.20                | ¥6,000  | \$37.20       |
| 2  | Metall parts for frame         | Custom made                | Ryouta Koudu / JLCPCB                    | <a href="https://jlcpcb.com/">https://jlcpcb.com/</a>                                                   | 1        | ¥5,000                     | \$31.00                | ¥5,000  | \$31.00       |
| 3  | Plastic parts for frame        | Custom made                | Ryouta Koudu                             | 3D Printer                                                                                              | 1        | ¥600                       | \$3.72                 | ¥600    | \$3.72        |
| 4  | Connecting parts               | Screws, nuts, washers, ... |                                          | Local tool store                                                                                        | 1        | ¥1,200                     | \$7.44                 | ¥1,200  | \$7.44        |
| 5  | Cables                         | Various cables             | Ryouta Koudu                             | <a href="https://www.jst-mfg.com/">https://www.jst-mfg.com/</a>                                         | 1        | ¥2,000                     | \$12.40                | ¥2,000  | \$12.40       |
| 6  | Wheels                         | Custom made                | Ryouta Koudu                             | <a href="http://www.engravingjapan.com/">http://www.engravingjapan.com/</a>                             | 4        | ¥100                       | \$0.62                 | ¥400    | \$2.48        |
| 7  | Drive motors                   | STS3032                    | Feetech RC MODEL CO., Ltd.               | <a href="https://www.feetechrc.com/">https://www.feetechrc.com/</a>                                     | 4        | ¥4,800                     | \$29.76                | ¥19,200 | \$119.04      |
| 8  | Motors for rescue kits         | MG90D                      | Tower Pro Pte Ltd                        | <a href="https://towerpro.com.tw/">https://towerpro.com.tw/</a>                                         | 2        | ¥1,350                     | \$8.37                 | ¥2,700  | \$16.74       |
| 9  | LED                            | WS2812B                    | Worldsemi Co.,Ltd                        | <a href="http://www.world-semi.com/">http://www.world-semi.com/</a>                                     | 31       | ¥25                        | \$0.16                 | ¥775    | \$4.81        |
| 10 | LCD Display                    | M154-240240-RGB            | SUNHOKEY Electronics Co.,Ltd.            | <a href="https://sunhokey.cn/">https://sunhokey.cn/</a>                                                 | 1        | ¥750                       | \$4.65                 | ¥750    | \$4.65        |
| 11 | Buzzer                         | UGCT7525AN4                | UNIVERSAL(CHANGZHOU) ELECTRONICS CO.,LTD | <a href="https://www.china-speakers-manufacturer.com/">https://www.china-speakers-manufacturer.com/</a> | 1        | ¥280                       | \$1.74                 | ¥280    | \$1.74        |
| 12 | Color sensor                   | S9706                      | Hamamatsu Photonics K.K.                 | <a href="https://www.hamamatsu.com/jp/en.html">https://www.hamamatsu.com/jp/en.html</a>                 | 1        | ¥600                       | \$3.72                 | ¥600    | \$3.72        |
| 13 | Reflectance sensor             | LBR127HLD                  | Letex Technology Corp.                   | <a href="https://www.letex.com.tw/index.aspx">https://www.letex.com.tw/index.aspx</a>                   | 2        | ¥80                        | \$0.50                 | ¥160    | \$0.99        |
| 14 | Switch                         | Various switches           | Cosland Co., Ltd.                        | <a href="https://www.cosland.com/about.php">https://www.cosland.com/about.php</a>                       | 1        | ¥500                       | \$3.10                 | ¥500    | \$3.10        |
| 15 | 9-axis IMU Sensor              | BNO085                     | Adafruit Industries                      | <a href="https://www.adafruit.com/">https://www.adafruit.com/</a>                                       | 1        | ¥4,074                     | \$25.26                | ¥4,074  | \$25.26       |
| 16 | Camera for victims             | UnitV M12 Camera           | M5Stack Technology Co., Ltd.             | <a href="https://m5stack.com/">https://m5stack.com/</a>                                                 | 2        | ¥7,282                     | \$45.15                | ¥14,564 | \$90.30       |
| 17 | Camera for color tiles         | Raspberry Pi Camera V3     | Raspberry Pi Foundation                  | <a href="https://www.raspberrypi.org/">https://www.raspberrypi.org/</a>                                 | 1        | ¥6,000                     | \$37.20                | ¥6,000  | \$37.20       |
| 18 | ToF Distance Sensor            | VL53L4CX                   | STMicroelectronics                       | <a href="https://www.st.com/content/st_com/en.html">https://www.st.com/content/st_com/en.html</a>       | 4        | ¥2,000                     | \$12.40                | ¥8,000  | \$49.60       |
| 19 | Single point LiDAR             | TSD10                      | PONO Technology Co.,Ltd.                 | <a href="https://positive-inno.com/">https://positive-inno.com/</a>                                     | 4        | ¥1,980                     | \$12.28                | ¥7,920  | \$49.10       |
| 20 | 360° LiDAR                     | T-mini Plus                | Shenzhen EAI Technology Co.,Ltd.         | <a href="https://www.ydlidar.com/">https://www.ydlidar.com/</a>                                         | 1        | ¥14,800                    | \$91.76                | ¥14,800 | \$91.76       |
| 21 | Single Board Computer          | Raspberry Pi 4 model B     | Raspberry Pi Foundation                  | <a href="https://www.raspberrypi.org/">https://www.raspberrypi.org/</a>                                 | 1        | ¥23,210                    | \$143.90               | ¥23,210 | \$143.90      |
| 22 | STM32 Microcontroller          | STM32F446RET6              | STMicroelectronics                       | <a href="https://www.st.com/content/st_com/en.html">https://www.st.com/content/st_com/en.html</a>       | 3        | ¥1,467                     | \$9.10                 | ¥4,401  | \$27.29       |
| 23 | STM32 Microcontroller          | STM32F103C8T6              | STMicroelectronics                       | <a href="https://www.st.com/content/st_com/en.html">https://www.st.com/content/st_com/en.html</a>       | 1        | ¥1,000                     | \$6.20                 | ¥1,000  | \$6.20        |
| 24 | Battery                        | LiPo 11V 1300mAh           | Kypom Technology Stock Co., Ltd          | <a href="http://kypom-battery.com/">http://kypom-battery.com/</a>                                       | 1        | ¥2,380                     | \$14.76                | ¥2,380  | \$14.76       |

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|----------------------------|---------------------------|--------------------------------|
| Japanese Yen JPY           | ¥0                        | \$0.00                         |

| #  | Name                | Software's Tool/Library | Description                               | Source                                                                                                                          | Autor               | Cost (Local Currency) | Cost (USA Dollars) |
|----|---------------------|-------------------------|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|---------------------|-----------------------|--------------------|
| 1  | Visual Studio Code  | Software's Tool         | Programmig (Raspberry Pi code)            | <a href="https://code.visualstudio.com/">https://code.visualstudio.com/</a>                                                     | Microsoft           | FREE                  | FREE               |
| 2  | Git, GitHub         | Software's Tool         | Version control for managing functions    | <a href="https://github.co.jp/">https://github.co.jp/</a>                                                                       | GitHub, Inc.        | FREE                  | FREE               |
| 3  | MaixPy IDE          | Software's Tool         | Programmig (STM32)                        | <a href="https://wiki.sipeed.com/maixpy/en/">https://wiki.sipeed.com/maixpy/en/</a>                                             | Sipeed              | FREE                  | FREE               |
| 4  | Autodesk Fusion     | Software's Tool         | 3D CAD shell design                       | <a href="https://www.autodesk.com/jp/products/fusion-360/overview">https://www.autodesk.com/jp/products/fusion-360/overview</a> | Autodesk Inc.       | FREE (Education)      | FREE (Education)   |
| 5  | KiCAD 9.0           | Software's Tool         | PCB design                                | <a href="https://www.kicad.org/">https://www.kicad.org/</a>                                                                     | OSS Contributors    | FREE                  | FREE               |
| 6  | Adafruit NeoPixel   | Library                 | For controlling WS2812B LEDs              | <a href="https://github.com/adafruit/Adafruit_NeoPixel">https://github.com/adafruit/Adafruit_NeoPixel</a>                       | Adafruit Industries | FREE                  | FREE               |
| 7  | Adafruit BNO08x     | Library                 | For controlling BNO085 IMU Sensor         | <a href="https://github.com/adafruit/Adafruit_BNO08x">https://github.com/adafruit/Adafruit_BNO08x</a>                           | Adafruit Industries | FREE                  | FREE               |
| 8  | Adafruit SSD1306    | Library                 | For controlling OLED display              | <a href="https://github.com/adafruit/Adafruit_SSD1306">https://github.com/adafruit/Adafruit_SSD1306</a>                         | Adafruit Industries | FREE                  | FREE               |
| 9  | STM32duino VL53L4CX | Library                 | For controlling VL53L4CX distance sensors | <a href="https://github.com/stm32duino/VL53L4CX">https://github.com/stm32duino/VL53L4CX</a>                                     | OSS Contributors    | FREE                  | FREE               |
| 10 | SCServo             | Library                 | For controlling STS3032 Servos            | <a href="https://github.com/workloads/scservo">https://github.com/workloads/scservo</a>                                         | Kerim Satirli       | FREE                  | FREE               |
| 11 | OpenCV              | Library                 | Image Processing                          | <a href="https://pypi.org/project/opencv-python/">https://pypi.org/project/opencv-python/</a>                                   | Foundation/OSS      | FREE                  | FREE               |
| 12 | picamera2           | Library                 | Image Processing                          | <a href="https://github.com/raspberrypi/picamera2">https://github.com/raspberrypi/picamera2</a>                                 | Raspberry Pi Ltd    | FREE                  | FREE               |
| 13 | Numpy               | Library                 | Array and math library                    | <a href="https://pypi.org/project/numpy/">https://pypi.org/project/numpy/</a>                                                   | OSS Contributors    | FREE                  | FREE               |
| 14 | ydliar              | Library                 | For controlling T-mini Plus LiDAR sensor  | <a href="https://github.com/YDLIDAR/YDLidar-SDK">https://github.com/YDLIDAR/YDLidar-SDK</a>                                     | Technology Co.,Ltd. | FREE                  | FREE               |
| 15 | Matplotlib          | Library                 | graphics generation                       | <a href="https://matplotlib.org/">https://matplotlib.org/</a>                                                                   | OSS Contributors    | FREE                  | FREE               |